

ISOLATION AND CHARACTERIZATION OF DOPAMINE D₂ RECEPTOR IN COPOLYMER-BASED NANODISCS

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ABSTRACT

The Dopamine D₂ receptor (D₂R) is an integral membrane protein belonging to the G-protein coupled receptor (GPCR) family. Embedded in the cell lipid membrane by its seven transmembrane helices, it is a major therapeutic target, antagonists being used to treat schizophrenia^{1,2} or agonists used in Parkinson's disease treatment². Isolation of membrane proteins, and GPCRs in particular, still represents a huge challenge, even if necessary for functional and structural studies. Among the different strategies used to isolate GPCRs, amphipathic copolymers represent an attractive option as they allow their extraction and purification in their native lipid environment, increasing the chances for retaining their function and structure³⁻⁵. The aim of this work was to obtain pure and functional D₂R surrounded by its native lipids in copolymer-based nanodiscs. A series of eight different copolymers was tested to screen for the best candidate for D₂R solubilization, purification and functionality. For all copolymers, the functionality was evaluated by a ligand binding assay using fluorescence polarization (FP). Specific ligand binding has been observed for DIBMA derivatives copolymers. In addition, D₂R basal activity was demonstrated by measuring GTPase activity of G_{i1} and G_{oA} heterotrimeric proteins, in DIBMA 12 nanodiscs.

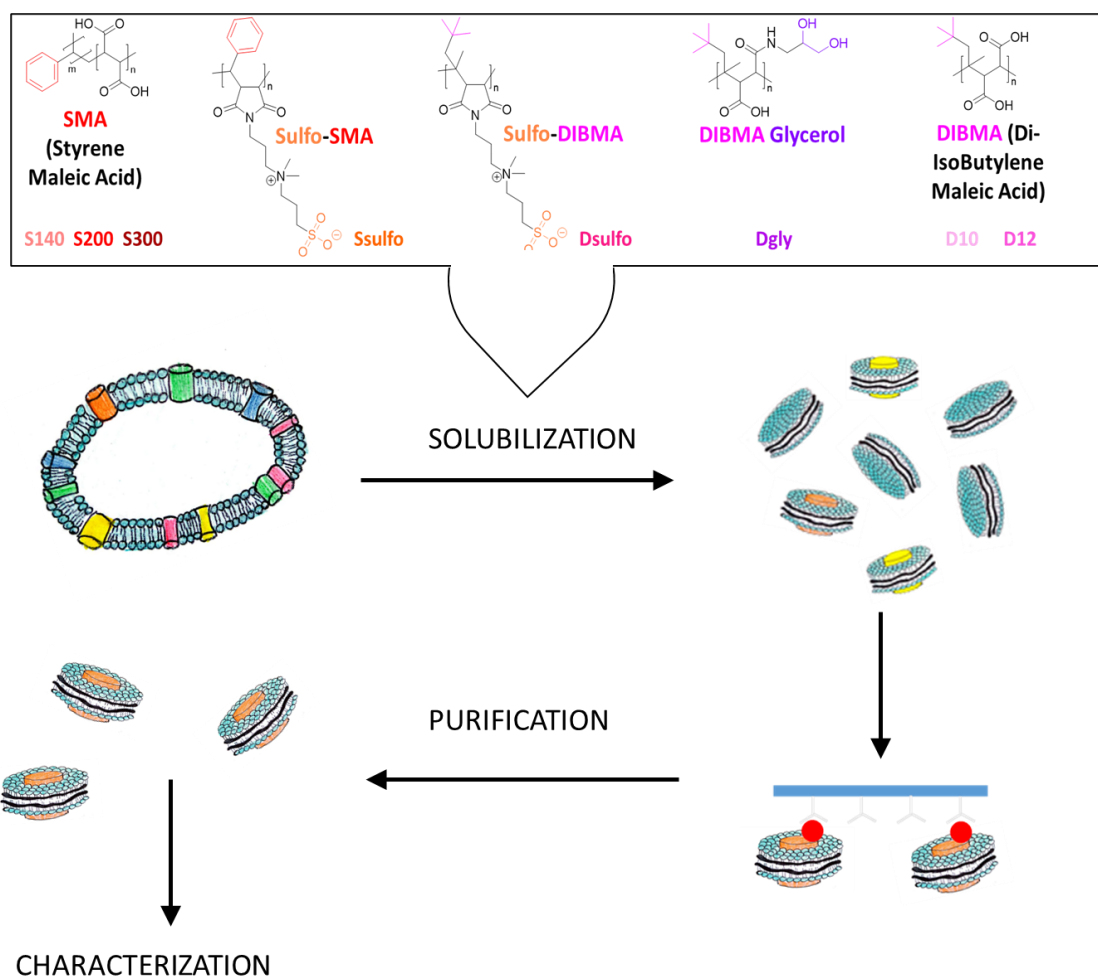


Fig. 8 different copolymers were used to compare their ability to solubilize cell membranes, purify the dopamine D2 receptor and maintain its functionality based on its ligand binding capacity and G protein activation

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